FOXA1/EZH2 Dual IHC Protocol Specification

Version 3.0 — Final

Antibody Selection, Staining Conditions, H-Score Calculation, Tissue Microarray Compatibility, Claudin-Low Scoring Warning, and Calibration Study Design

Supporting document for CS-LIT-1 (DOI: 10.5281/zenodo.18883922) and CS-LIT-22 (DOI: 10.5281/zenodo.18892788)

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March 6, 2026

VERSION HISTORY:

- **v1.0 (2026-03-06):** Initial release. Antibody selection, staining protocol, H-score method, calibration study design.
- **v2.0 (2026-03-06):** Added IHC/MS dynamic range rationale (Section 2). Added claudin-low stromal contamination warning (Section 8). Revised cut-point table to RNA-space framing with red critical box (Section 9).
- **v3.0 (2026-03-06):** Added tissue microarray (TMA) compatibility section (Section 4.2). Final version for distribution.

STATUS: Pre-calibration protocol specification. Antibody recommendations and staining conditions are based on published breast cancer IHC literature and manufacturer datasheets. **No IHC H-score cut-points have been established.** The RNA-space ratio values from computational validation (LumA 9.38, LumB 8.10, HER2 3.34, TNBC 0.52, CL 0.10) are not transferable to IHC H-score units. See Section 9 for the correct interpretation and Section 10 for the calibration study design.

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1 Purpose and Scope

This document provides the complete technical specification for running the FOXA1/EZH2 dual IHC ratio assay on formalin-fixed, paraffin-embedded (FFPE) breast cancer tissue, in both whole-section and tissue microarray (TMA) formats. It is intended for:

- Pathologists and histotechnologists implementing the assay in a clinical or research setting
- Researchers designing the IHC concordance study described in CS-LIT-22
- Institutions evaluating the assay for inclusion in a breast cancer diagnostic panel
- Institutions whose archival tissue is held in TMA format

This document has been revised to incorporate all relevant findings and limitations from the OrganismCore master computational artifact (FOXA1-EZH2-RATIO-MASTER-RA, 2026-03-06), which records the complete four-script validation across ∼7,500 patients in seven independent datasets. Three critical points from that artifact drive this protocol:

1. The protein signal is confirmed at the protein level by CPTAC mass spectrometry ($r = -0.492$, $n = 122$). The MS kappa failure was a dynamic range artefact, not a biology failure. IHC has 800× more dynamic range than iTRAQ MS.
2. The claudin-low stromal contamination artefact observed in bulk RNA must be prevented in IHC by disciplined cancer-cell-only scoring.
3. IHC H-score cut-points do not yet exist. RNA-space values must not be used as IHC thresholds.

2 Why IHC Is Expected to Outperform Mass Spectrometry: The Dynamic Range Argument

2.1 What CPTAC mass spectrometry showed

The CPTAC-BRCA iTRAQ proteomics dataset ($n = 122$) confirmed:

- Correct protein-level subtype ordering: LumA > LumB > HER2 > TNBC
- Strong inverse correlation: Spearman $r = -0.492$ between FOXA1 and EZH2 protein across all 122 patients ($p < 0.0001$)
- Kappa-based subtype classification: **failed**

2.2 Why the kappa failed and why this is irrelevant to IHC

THE DYNAMIC RANGE ARGUMENT:

iTRAQ MS compressed all four breast cancer subtypes into **0.38 $\log_2$ units** total range (LumA: −0.320, TNBC: −0.733). Measurement noise at this scale exceeds subtype signal. Threshold-based classification is mathematically impossible at 0.38 $\log_2$ units of separation.

IHC H-scores span **0–300** — providing approximately **800**× more dynamic range. A LumA tumour with strong FOXA1 (H-score ≈ 240) and low EZH2 (H-score ≈ 30) gives a ratio of ≈ 8.0. A TNBC tumour with absent FOXA1 (H-score ≈ 10) and high EZH2 (H-score ≈ 210) gives a ratio of ≈ 0.05. The separation that is invisible at 0.38 MS units is large and classifiable at IHC scale.

The MS kappa failure is a reason to be more confident in IHC, not less. The protein signal exists and is directionally correct. The instrument used to measure it (MS) lacked the range to classify. IHC has the range.

3 Antibody Specifications

3.1 FOXA1 Antibody

Field	Specification
Target protein	FOXA1 (Forkhead Box A1)
Recommended clone	**D4E2 (XP Rabbit Monoclonal)**
Manufacturer	Cell Signaling Technology
Catalogue number	CST #53528
Host / Clonality	Rabbit monoclonal
Cellular localisation	Nuclear exclusively
Validated for FFPE IHC	Yes (manufacturer validated)
Manufacturer datasheet	cellsignal.com #53528
Alternative clone	SP87 (Rabbit Monoclonal), Abcam ab170933; equivalent published performance in breast cancer IHC
Biological role	Pioneer transcription factor defining luminal identity. High FOXA1 = luminal programme open and active. Low FOXA1 = luminal identity lost or epigenetically silenced.

3.2 EZH2 Antibody

Field	Specification
Target protein	EZH2 (Enhancer of Zeste Homolog 2)
Recommended clone	**D2H1 (XP Rabbit Monoclonal)**
Manufacturer	Cell Signaling Technology
Catalogue number	CST #5246
Host / Clonality	Rabbit monoclonal
Cellular localisation	Nuclear exclusively
Validated for FFPE IHC	Yes (manufacturer validated)
Manufacturer datasheet	cellsignal.com #5246
Alternative clone	AC22 (Mouse Monoclonal), Leica Biosystems NCL-L-EZH2; widely published in breast cancer; requires anti-mouse secondary
Biological role	PRC2 catalytic subunit. Deposits H3K27me3 silencing marks on FOXA1, GATA3, ESR1 promoters. High EZH2 = active epigenetic suppression of luminal identity.

ANTIBODY MATCHING NOTE: Both D4E2 (FOXA1) and D2H1 (EZH2) are rabbit monoclonal antibodies from the same manufacturer. The same anti-rabbit HRP polymer secondary can be used for both stains when run sequentially. Both produce exclusively nuclear staining, restricting H-score calculation to the nuclear compartment and eliminating cytoplasmic staining ambiguity.

4 Tissue Section Preparation — Whole Section and TMA

4.1 Whole-section FFPE specimens

- **Section thickness:** 3–4 μm
- **Sections required per case:** Two serial or near-serial sections (one for FOXA1, one for EZH2), within 10 levels to ensure the same tumour area is represented in both stains
- **Mounting:** Charged or poly-L-lysine coated slides
- **Drying:** 60°C oven, minimum 60 minutes
- **Tumour content:** Select sections with $\geq 20\%$ invasive tumour cells. A pathologist must mark the invasive tumour area on the H&E before IHC scoring.
- **FFPE block age:** Any age is suitable. IHC does not require RNA integrity — a key advantage over PAM50, which requires adequate RNA and is unreliable on archival or poorly fixed tissue.

4.2 Tissue microarray (TMA) specimens

TMA COMPATIBILITY:

This protocol applies without modification to tissue microarray format. TMAs are the standard infrastructure for retrospective IHC validation studies at scale in major cancer centres. Institutions whose archival breast cancer tissue is held in TMA format are directly compatible with this protocol. No format conversion is required.

TMA-specific requirements:

- **Minimum core diameter:** 1.0 mm. Cores of 1.5–2.0 mm are preferred for adequate tumour cell count.
- **Minimum cores per case:** Two cores per case per stain are recommended. Cases with only one evaluable core per stain should be flagged and reported separately in the analysis.
- **Scoring on TMA cores:** Apply the same H-score method (Section 7) to the evaluable tumour area within each core. Score invasive cancer cell nuclei only; exclude stroma, lymphocytes, and normal epithelium within the core.
- **Two-core averaging:** Where two cores are available for a case, compute the H-score ratio for each core independently and report the mean ratio as the case value. If the two core ratios differ by more than 0.5 log units, flag for pathologist review (may indicate tumour heterogeneity or core placement in stroma rather than tumour).
- **CL cores specifically:** The claudin-low stromal contamination warning (Section 8) applies with equal force to TMA cores. A 1.0 mm core punched into a mesenchymal-rich CL tumour may have low cancer cell density. If < 50 invasive cancer cell nuclei are identifiable in the core, exclude the case from the primary analysis and report it as unevaluable.
- **Tissue loss:** TMA cores are susceptible to section loss on sequential cutting. For each TMA block, cut at minimum three sequential sections at the start of the run to ensure paired FOXA1 and EZH2 sections are available for each core.

TMA advantage for this study: A TMA containing 400 cases requires a single pair of slides per staining run rather than 400 individual slides. This reduces reagent cost, staining variability between cases (all cases processed under identical conditions simultaneously), and pathologist scoring time (one slide field rather than 400 individual section scans). For institutions with archival TMA infrastructure, this is the recommended format for the calibration study.

5 Staining Protocol

5.1 Manual protocol

Step	Procedure	Conditions / Notes
1	Deparaffinise	Xylene 3 × 5 min or equivalent
2	Rehydrate	Graded ethanol 100%, 95%, 70%, distilled water, 2 min each
3	Endogenous peroxidase block	3% H_2O_2 in methanol or water, 10 min RT. Wash 3× PBS.
4a	**Antigen retrieval (FOXA1 sections)**	**HIER, Citrate buffer pH 6.0.** Pressure cooker or steamer, 20 min at 95–100°C. Cool 20–30 min RT. Wash 2× PBS.
4b	**Antigen retrieval (EZH2 sections)**	**HIER, EDTA buffer pH 9.0.** Pressure cooker or steamer, 20 min at 98°C. Cool 20–30 min RT. Wash 2× PBS. *Run FOXA1 and EZH2 sections in separate retrieval runs owing to different buffer requirements.*
5	Serum block	5% normal goat serum in PBS, 30 min RT. Tip off; do not wash.
6a	**Primary antibody (FOXA1)**	**D4E2 (CST #53528) 1:100 in antibody diluent.** 60 min RT, humidified chamber. Wash 3× PBS, 5 min each.
6b	**Primary antibody (EZH2)**	**D2H1 (CST #5246) 1:100 in antibody diluent.** 60 min RT, humidified chamber. Wash 3× PBS, 5 min each.
7	Secondary antibody	Anti-rabbit HRP-polymer (e.g. Dako EnVision+ K4003 or equivalent). 30 min RT. Wash 3× PBS.
8	Chromogen development	DAB substrate-chromogen solution. 3–8 min RT; monitor microscopically. Stop when brown nuclear staining is visible with minimal background.
9	Counterstain	Mayer's haematoxylin 2 min. Blueing in running tap water 5 min.
10	Dehydrate and mount	Graded ethanol, xylene, permanent mounting medium.

5.2 Automated platform adaptations

Platform	FOXA1 (D4E2)	EZH2 (D2H1)
Ventana BenchMark (Roche)	CC1 (Tris-EDTA pH 8.5), 36 min 100°C. D4E2 1:100. OptiView DAB.	CC1, 36 min 100°C. D2H1 1:100. OptiView DAB.
Dako AutoStainer (Agilent)	EnVision Flex low pH (citrate), 20 min 98°C. D4E2 1:100.	EnVision Flex high pH (EDTA), 20 min 98°C. D2H1 1:100.
Leica Bond RX	ER1 (citrate pH 6), 20 min. D4E2 1:100. Bond Polymer Refine.	ER2 (EDTA pH 9), 20 min. D2H1 1:100. Bond Polymer Refine.

6 Tissue Controls

Control type	FOXA1	EZH2
Positive tissue	LumA breast carcinoma (known ER+, LumA by PAM50 or clinical behaviour). Strong diffuse nuclear staining expected.	TNBC or basal-like breast carcinoma. Strong diffuse nuclear staining expected.
Negative tissue	Claudin-low or TNBC. Absent or near-absent nuclear staining expected. *Score cancer cells only — see Section 8.*	Normal breast stroma. Weak or absent nuclear staining expected.
Internal positive	Normal luminal ductal epithelium on same section. Moderate FOXA1 nuclear staining.	Proliferating inflammatory cells (process control only; do not use for scoring calibration).
Negative reagent	Omit primary; isotype-matched IgG at same concentration.	Omit primary; isotype-matched IgG at same concentration.

7 Scoring Method

7.1 What to score

SCORE INVASIVE CANCER CELL NUCLEI ONLY.

Exclude without exception:

- Normal ductal and lobular epithelium
- Stromal fibroblasts and endothelial cells
- Tumour-infiltrating lymphocytes
- Myoepithelial cells
- Necrotic areas
- Any non-cancer nuclear staining

A pathologist must mark the invasive tumour area on the H&E before IHC scoring. Score the marked area only.

This instruction applies to all subtypes. It is especially critical for claudin-low. See Section 8.

7.2 H-score calculation

$$\text{H-score} = (1 \times p_1) + (2 \times p_2) + (3 \times p_3) \tag{1}$$

where p_1, p_2, p_3 are the percentages of invasive cancer cell nuclei with weak (1+), moderate (2+), and strong (3+) staining respectively. Estimate to the nearest 5%. **H-score range: 0–300.**

Intensity	Visual description	Practical example
0 (negative)	No nuclear staining above background	FOXA1 in claudin-low
1+ (weak)	Light brown nuclear tinge; faint but identifiable; above background	FOXA1 in TNBC
2+ (moderate)	Clearly brown; unambiguous; not intense	FOXA1 in LumB; EZH2 in LumA
3+ (strong)	Intense brown; visible at low power	FOXA1 in LumA; EZH2 in TNBC

7.3 Ratio calculation

$$\text{FOXA1/EZH2 ratio} = \frac{\text{FOXA1 H-score}}{\text{EZH2 H-score}} \tag{2}$$

Division by zero: If EZH2 H-score = 0, record ratio as undefined and classify as claudin-low by default. Document these cases separately in the calibration dataset.

7.4 Worked example

TNBC section:

- FOXA1: 5% at 1+, 2% at 2+, 0% at 3+ $\Rightarrow$ H-score $= 5 + 4 + 0 = 9$
- EZH2: 20% at 1+, 45% at 2+, 30% at 3+ $\Rightarrow$ H-score $= 20 + 90 + 90 = 200$
- Ratio $= 9 \div 200 = 0.045$

This falls in the low-ratio TNBC/CL zone, consistent with the framework.

7.5 Minimum cell count

Score a minimum of **200 invasive cancer cell nuclei** per stain on whole sections. For TMA cores, score all evaluable invasive nuclei; flag cores with < 50 evaluable nuclei as unevaluable.

7.6 Inter-observer assessment

Each slide must be independently scored by two pathologists, blinded to PAM50 subtype and to each other's score. Report:

- ICC for continuous H-score values; target > 0.80
- Cohen's κ for categorical subtype assignment; target > 0.70

8 Claudin-Low Stromal Contamination Warning

CRITICAL SCORING WARNING: CLAUDIN-LOW TUMOURS

This is the single most consequential scoring error possible in this assay.

The problem:

Claudin-low (CL) breast tumours are highly mesenchymal, with abundant stromal content relative to cancer cell density. That stroma consists predominantly of EZH2-low mesenchymal cells — fibroblasts, myofibroblasts, vascular endothelium. These stromal cells are EZH2-negative or EZH2-low by IHC.

If stromal cell nuclei are included in the EZH2 count, the EZH2 H-score is artificially lowered. This falsely elevates the FOXA1/EZH2 ratio, causing a CL tumour that should score at the lowest ratio to appear in the luminal range — the exact opposite of the correct classification.

This artefact was directly observed and documented.

In bulk RNA sequencing data (which averages cancer cell signal together with stromal signal), claudin-low tumours appeared **above LumA** in the ratio ordering — placing CL as the most luminal-appearing subtype, when it is biologically the least luminal. The mechanism: bulk RNA captures EZH2-low mesenchymal stroma, artificially lowering the denominator and inflating the ratio. This is recorded in the OrganismCore master computational artifact (result R4-C, CL ratio = 1.4888 in METABRIC bulk RNA vs CL ratio = 0.10 in scRNA-seq cancer cells only).

IHC avoids this artefact **only if** the scorer counts cancer cell nuclei exclusively.

The instruction:

For every case, and especially for suspected CL or mesenchymal tumours:

1. Score only nuclei within the marked invasive tumour area
2. Exclude all stromal nuclei, even those immediately adjacent to or interspersed with cancer cells
3. Exclude all TIL nuclei within tumour nests
4. When uncertain whether a nucleus belongs to a cancer cell or a stromal cell, exclude it
5. Flag all cases scored in a predominantly mesenchymal background for independent pathologist review before classification

The self-check for correct CL scoring:

In a correctly scored CL case:

- FOXA1 H-score should be very low to absent (CL FOXA1 is −97.8% below normal LumA in single-cell data)
- EZH2 H-score should be moderate, not zero and not high (CL EZH2 is elevated +67% above normal but not at the TNBC level of +189%)
- The ratio should therefore be very low: low numerator divided by moderate denominator

If a scored CL case gives a ratio in the luminal range, stromal contamination of the EZH2 count is the most likely cause. Re-score cancer cell nuclei exclusively.

TMA note: A 1.0 mm TMA core punched into a mesenchymal-rich CL tumour may have low cancer cell density. If < 50 invasive cancer cell nuclei are identifiable, exclude the case from the primary analysis and report as unevaluable.

9 Provisional Ratio Ranges and Their Correct Interpretation

CRITICAL: THESE ARE NOT IHC CUT-POINTS.

The values below are **single-cell RNA-seq ratio values** in mean expression units. They are not IHC H-score values. There is no formula to convert between RNA-space and IHC H-score space. They depend on entirely different measurement principles, instruments, and calibration conditions.

Do not use these values as starting- point thresholds for the calibration study.

These values indicate three things only:

1. **Rank order:** LumA highest, CL lowest. This rank order is expected to hold in IHC.
2. **Span:** Two orders of magnitude from LumA (9.38) to CL (0.10). This suggests IHC will also show large separation, even if absolute values differ.
3. **Direction:** High ratio → luminal. Low ratio → undifferentiated/epigenetically locked.

IHC H-score cut-points must be determined in the calibration study using Youden J index optimisation from ROC analysis against PAM50 ground truth. See Section 10.

Subtype	scRNA-seq ratio (RNA units only)	Expected IHC character	Treatment logic
Luminal A	9.38	High FOXA1 H-score. Low EZH2 H-score. Ratio: high (exact value TBD).	CDK4/6 inhibitor + ET.
Luminal B	8.10	Moderate-high FOXA1. Low-moderate EZH2. Ratio: below LumA (TBD).	HDACi + ET. CDK4/6i also active.
HER2-enriched	3.34	Moderate FOXA1. Moderate EZH2. Ratio: mid-range (TBD).	Anti-HER2 first. ET thereafter.
TNBC	0.52	Low FOXA1. High EZH2. Ratio: low (TBD).	Tazemetostat → fulvestrant.
Claudin-low	0.10	Absent/very low FOXA1. Moderate EZH2 (*cancer cells only; see Section 8*). Ratio: lowest (TBD).	Anti-TIGIT → anti-PD-1 sequence.
ILC (exception)	Above LumA range	FOXA1 *hyperactivated* above normal LumA level. EZH2 low. CDH1 absent.	Fulvestrant > AI. Full ER degradation.

ILC identification: Three features together: (1) FOXA1 H-score above normal LumA level; (2) CDH1-absent by IHC or morphological confirmation of lobular histology; (3) single-file infiltration pattern on H&E. CDH1 IHC may be added as a third stain when lobular vs ductal distinction is required.

10 Calibration Study Design

10.1 Purpose

The calibration study must determine IHC H-score cut-points from scratch, using actual IHC H-scores against PAM50 ground truth. It must not anchor on RNA-space values.

10.2 Study type

Retrospective IHC concordance study. No new patient recruitment. No treatment changes. Observational only. Compatible with both whole-section and TMA formats.

10.3 Study population

Archived FFPE breast cancer tissue with:

- Available PAM50/Prosigna result
- Confirmed invasive breast carcinoma
- Adequate tumour content ($\geq$ 20% invasive tumour cells on whole section; $\geq$ 50 evaluable cancer cell nuclei on TMA core)

Target sample size:

PAM50 subtype	Target n	Minimum
Luminal A	80–100	50
Luminal B	80–100	50
HER2-enriched	50–60	30
Basal/TNBC	60–80	40
Claudin-low	30–40	20
Total	**300–380**	**190**

Include a minimum of 30 ILC cases (CDH1-absent) as a separate analysis arm.

10.4 Primary endpoint

Agreement between FOXA1/EZH2 IHC ratio classification and PAM50:

- Cohen's $\kappa \geq 0.70$
- Overall concordance $\geq$ 75%
- AUC for LumA vs Basal and LumA vs LumB ≥ 0.75

10.5 Cut-point determination method

REQUIRED METHOD:

1. Split cohort into training (70%) and test (30%) sets, stratified by PAM50 subtype.
2. In training set: run ROC analysis for each binary task (LumA vs LumB, LumA vs Basal, TNBC vs CL). Determine optimal H-score ratio threshold per boundary using Youden J index ($\max(\text{sensitivity} + \text{specificity} - 1)$).
3. Validate cut-points in held-out test set. Report sensitivity, specificity, PPV, NPV, κ.
4. **Do not use RNA-space values (9.38, 8.10, 3.34, 0.52, 0.10) as anchors at any step.**
5. Report cut-points with 95% CI from bootstrapping ($\geq$ 1,000 iterations).

10.6 Secondary endpoints

- Inter-observer reliability (ICC and κ between two pathologists)
- Correlation between IHC ratio and clinical outcomes where available (exploratory)
- Confirmation that CL scores below TNBC in IHC (pre-specified hypothesis; CL stromal contamination check as sensitivity analysis)
- TMA vs whole-section concordance (where both formats are available for the same cases)

10.7 Estimated resources

Resource	Estimate
Antibody cost	FOXA1 D4E2 #53528: ∼$400/100 tests. EZH2 D2H1 #5246: ∼$400/100 tests. For 400 cases: ∼$1,500–$3,000.
IHC consumables	∼$2,000–$4,000.
Pathologist time	∼40–60 hours (2 pathologists, 400 cases, 2 stains). TMA format reduces this substantially.
Statistical analysis	∼10–20 hours.
Total (consumables)	**$5,000–$15,000**
Timeline	**3–6 months**

11 Troubleshooting

Problem	Likely cause	Solution
No or very weak staining (both antibodies)	Antigen retrieval failure; antibody inactivation; poor section adhesion	Increase retrieval time. Check antibody storage (−20°C). Use freshly cut sections (< 6 weeks).
High EZH2 background	Insufficient peroxidase block; non-specific binding	Extend H_2O_2 block to 15 min. Increase serum block concentration.
FOXA1 cytoplasmic staining	Non-specific binding	Score nuclear staining only. Cytoplasmic signal excluded.
CL ratio in luminal range	**Stromal contamination of EZH2 count — dominant error.**	**Re-score cancer cell nuclei exclusively. Exclude all stromal nuclei. See Section 8.**
TMA core loss on serial sectioning	Inadequate cutting technique; core detachment	Cut ≥ 3 sequential sections at run start. Use adhesive-coated slides for TMA.
Poor section quality	Block quality; section age; inadequate drying	Re-cut from block. Use sections < 2 weeks old.

12 Document Status and Contact

Field	Value
Document	IHC Protocol Specification v3.0 (Final)
Status	Pre-calibration. No IHC cut-points established.
Author	Eric Robert Lawson, OrganismCore
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ORCID	0009-0002-0414-6544
CS-LIT-1 DOI	10.5281/zenodo.18883922
CS-LIT-22 DOI	10.5281/zenodo.18892788
Repository	Eric-Robert-Lawson/attractor-oncology
Master artifact	FOXA1-EZH2-RATIO-MASTER-RA, 2026-03-06
Version history	v1.0: Initial release. v2.0: CL warning; cut-point reframe; dynamic range rationale. v3.0
Next version trigger	Calibration study completion; update to validated IHC cut-points.